

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

***CO3:** Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I EDRLN J S (192524041) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: edrlnjs

Cloud Computing Case Studies: Architecture Component Analysis

Each case study below is broken down into its core cloud architecture components — Clients, APIs / Web Services, Cloud Storage, Platform Services, Network Infrastructure, and Security — based on the scenario described.

Case Study 1: Smart Retail Inventory Management System

Clients	POS (Point-of-Sale) systems and analytics dashboards used by store staff and managers to interact with the inventory system.
APIs / Web Services	RESTful APIs connect store devices (clients) to centralized cloud services, enabling communication between POS systems and the backend.
Cloud Storage	Stores product data, transaction logs, and analytics datasets generated across the retail chain.
Platform Services (PaaS)	Managed analytics and demand-forecasting services process the stored data to generate business insights.
Network Infrastructure	Connects distributed store locations to centralized cloud services, enabling real-time data flow.
Security	VPNs, firewalls, and API gateways protect access between stores and the cloud, ensuring only authorized systems can communicate.

Summary: This case study demonstrates a full cloud stack — clients (POS/dashboards) → APIs → cloud storage → platform (analytics/forecasting) services, secured through VPNs, firewalls, and API gateways — integrating infrastructure, platform services, and web applications for efficient retail operations.

Case Study 2: Cloud-Based Document Collaboration System

Clients	Web browsers used by end users to access the document collaboration platform (similar to Google Docs) for real-time editing and sharing.
APIs / Web Services	Web APIs enable synchronization of document changes across multiple clients and devices in real time.
Cloud Storage	Distributed cloud storage holds documents and maintains version control/history so changes can be tracked and reverted.
Platform Services (PaaS)	Underlying platform manages real-time collaboration logic, synchronization, and version control mechanisms.
Network Infrastructure	Global network infrastructure ensures low latency and high availability so users anywhere can collaborate smoothly.
Security	Encryption, identity management, and access permissions control who can view, edit, or share each document.

Summary: This case study centers on browser-based clients synchronized via web APIs to distributed, version-controlled cloud storage, supported by a low-latency global network and secured through encryption, identity management, and permissions.

Case Study 3: Online Food Ordering Platform

Clients	Web browsers and mobile apps used by customers to browse menus and place orders.
APIs / Web Services	RESTful Web APIs hosted on the cloud platform connect frontend clients to backend services (orders, payments, restaurant management).
Cloud Storage	Securely stores menu images, invoices, and customer data.

Platform Services / Infrastructure (IaaS/PaaS)	Virtual machines and managed Kubernetes provide scalable, reliable compute infrastructure to run backend services.
Network Infrastructure	Cloud-hosted network connects mobile/web clients to backend services via HTTPS, supporting scalability as order volume grows.
Security	Authentication tokens, HTTPS, and role-based access control (RBAC) protect customer data and restrict access based on user roles.

Summary: This case study shows a mobile/web client base talking to RESTful APIs on a cloud platform, backed by VM- and Kubernetes-based infrastructure for scalability, with cloud storage for data and strong security via tokens, HTTPS, and RBAC — demonstrating integration of clients, APIs, storage, and platform services in real time.

Cross-Case Comparison

Common Pattern	All three systems follow the same layered cloud architecture: Client → API/Web Service → Backend/Platform Service → Cloud Storage, wrapped in a secured network.
Scalability Approach	Case 1 scales via managed analytics services; Case 2 scales via distributed storage and global network reach; Case 3 scales via VMs and Kubernetes orchestration.
Security Approach	Case 1 uses VPNs/firewalls/API gateways; Case 2 uses encryption/identity management; Case 3 uses tokens/HTTPS/RBAC — each tailored to its risk profile (retail data, sensitive documents, customer PII/payments).